

Density-Aware Geodesic Distance Definition Memo

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This memo is Deliverable 2 for the density-aware geodesic-distance research track. It fixes the working gflow vocabulary for density-aware distances. The companion literature map covers history, sources, and adoption; this memo is more axiomatic. Its goal is to say what objects gflow should mean when it says density-aware geodesic, graph distance, Fermat distance, diffusion distance, PHATE potential distance, or oracle truth.

1. Notation

Let

$$(M, g)$$

be a connected smooth Riemannian manifold, possibly with boundary, with arclength element ds_g . Let

$$\rho : M \rightarrow [0, \infty)$$

denote a sampling, support, or biological-state density. In finite data, let

$$X = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$$

or more generally $X \subset M$. A graph on X is

$$G = (V, E), \quad V = \{1, \dots, n\}.$$

For an edge $e = (i, j)$, let $\ell_{ij} \geq 0$ be its base geometric length, usually

$$\ell_{ij} = \|x_i - x_j\|$$

after the chosen preprocessing. Let $\hat{\rho}_i$ be an estimated vertex density and $\hat{\rho}_{ij}$ an edge density summary.

Throughout this memo, $\alpha \geq 0$ is the gflow density penalty exponent: larger α makes low-density regions more expensive. This is not the same parameter as diffusion-map alpha normalization. When a diffusion-map normalization exponent is needed, write it as a .

2. Basic Distance Vocabulary

A function $d : X \times X \rightarrow [0, \infty]$ is an extended metric if it is nonnegative, symmetric, zero exactly on the diagonal, satisfies the triangle inequality, and may take value ∞ . It is a pseudometric if distinct points may have distance zero. It is a directed distance or quasi-metric if the triangle inequality holds but symmetry may fail. It is a graph metric when it is the shortest-path distance induced by nonnegative edge weights on an undirected connected graph.

This memo reserves the phrase **geodesic distance** for:

1. a continuous path metric obtained by minimizing a path length over curves;

2. a graph shortest-path metric intended to estimate such a continuous path metric;
3. an oracle path metric on a known synthetic support.

Diffusion, PHATE, commute, resistance, heat-kernel, and PageRank distances are support-aware or random-walk distances, but should not be called geodesic distances unless a specific small-time or shortest-path limit is being invoked.

3. Continuous Density-Weighted Geodesic

Definition

For $\alpha \geq 0$, define the density-weighted length of a piecewise C^1 curve $\gamma : [0, 1] \rightarrow M$ by

$$L_{\rho, \alpha}(\gamma) = \int_0^1 \rho(\gamma(t))^{-\alpha} \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))} dt.$$

The associated density-weighted distance is

$$d_{\rho, \alpha}(x, y) = \inf_{\gamma(0)=x, \gamma(1)=y} L_{\rho, \alpha}(\gamma).$$

When $\alpha = 0$, this reduces to the ordinary Riemannian geodesic distance on (M, g) .

Conformal Metric Interpretation

If ρ is positive and smooth, then $d_{\rho, \alpha}$ is the geodesic distance of the conformally transformed Riemannian metric

$$g_{\rho, \alpha} = \rho^{-2\alpha} g.$$

The factor of two appears because path length uses the square root of the metric tensor:

$$\sqrt{g_{\rho, \alpha}(\gamma', \gamma')} = \rho^{-\alpha} \sqrt{g(\gamma', \gamma')}.$$

Metric Status

If M is connected and ρ is positive and finite everywhere, then $d_{\rho, \alpha}$ is a metric on M . If ρ vanishes on barriers, then distances crossing those barriers may be infinite; this is an extended metric on each reachable support component. If ρ is unbounded, distances between distinct points can collapse toward zero in degenerate cases; gflow should avoid this by working with bounded or transformed densities.

Practical Robust Version

For finite samples, use a transformed density

$$\rho^*(x) = T(\hat{\rho}(x)),$$

where T is a monotone positive transform. Important examples are:

$$\rho^*(x) = \max\{\widehat{\rho}(x), \rho_{\min}\},$$

$$\rho^*(x) = \min\{\max\{\widehat{\rho}(x), \rho_{\min}\}, \rho_{\max}\},$$

and a penalty cap

$$\rho^*(x)^{-\alpha} = \min\{\widehat{\rho}(x)^{-\alpha}, c_{\max}\}.$$

These produce a metric for the transformed density. Diagnostics must report how much of the data or edge set is controlled by the floor or cap.

4. Graph Density-Weighted Geodesic

Definition

Let $G = (V, E)$ be an undirected graph. For each edge $(i, j) \in E$, define

$$w_{ij} = \frac{\ell_{ij}}{\phi_{ij}^\alpha}, \quad \phi_{ij} > 0,$$

where ϕ_{ij} summarizes density along or near the edge. The graph density-weighted distance is

$$d_{G, \phi, \alpha}(i, j) = \min_{\pi: i \rightarrow j} \sum_{(u, v) \in \pi} w_{uv}.$$

Edge Density Summaries

Candidate summaries include

$$\phi_{ij}^{\text{arith}} = \frac{\widehat{\rho}_i + \widehat{\rho}_j}{2},$$

$$\phi_{ij}^{\text{geom}} = \sqrt{\widehat{\rho}_i \widehat{\rho}_j},$$

$$\phi_{ij}^{\text{harm}} = \frac{2}{\widehat{\rho}_i^{-1} + \widehat{\rho}_j^{-1}},$$

$$\phi_{ij}^{\min} = \min\{\widehat{\rho}_i, \widehat{\rho}_j\},$$

and, when coordinates and an estimator are available,

$$\phi_{ij}^{\text{mid}} = \widehat{\rho}\left(\frac{x_i + x_j}{2}\right).$$

The minimum and harmonic summaries penalize low-density endpoints more strongly. Midpoint or edge-neighborhood summaries are better for detecting edges that cross gaps between two dense endpoint regions, but they require a meaningful ambient or latent coordinate system.

Metric Status

If G is undirected, connected, and all $w_{ij} > 0$, then $d_{G,\phi,\alpha}$ is a metric on vertices. If G is disconnected, it is an extended metric with infinite distances between components. If zero-weight edges are allowed, the result is a pseudometric. If $w_{ij} \neq w_{ji}$ or the graph is directed, the shortest-path distance is a directed distance, not a metric.

Relationship to the Continuous Metric

The graph distance estimates $d_{\rho,\alpha}$ only when:

1. the graph topology is locally consistent with the intended support;
2. ℓ_{ij} estimates local arclength at the edge scale;
3. ϕ_{ij} estimates the density encountered along the edge;
4. graph resolution increases while edge lengths shrink.

Edge reweighting cannot fully fix a graph that contains unsupported shortcuts unless the edge density summary sees the low-density gap crossed by the edge.

5. Fermat / Power-Weighted Sample Distance

Rooted Convention

For $p \geq 1$, define

$$\ell_p(x, y; X) = \min_{x=x_0, \dots, x_m=y} \left(\sum_{r=0}^{m-1} \|x_{r+1} - x_r\|^p \right)^{1/p}.$$

Unrooted Convention

Many topology papers use

$$d_{X,p}(x, y) = \min_{x=x_0, \dots, x_m=y} \sum_{r=0}^{m-1} \|x_{r+1} - x_r\|^p.$$

These are related by $d_{X,p} = \ell_p^p$ when the same admissible graph or complete point cloud is used.

Continuum Correspondence

For samples from an intrinsic d -dimensional density ρ , the continuum Fermat geodesic has integrand

$$\rho^{-(p-1)/d} ds_g.$$

Thus the gflow correspondence is

$$\alpha = \frac{p-1}{d}, \quad p = 1 + \alpha d.$$

The corresponding conformal metric tensor is

$$g_p = \rho^{-2(p-1)/d} g.$$

Metric Status

On a connected graph with positive Euclidean edge lengths, ℓ_p is a metric for $p \geq 1$. The unrooted $d_{X,p}$ is also a path metric because it is a shortest-path distance with edge weights $\|x_i - x_j\|^p$. In the complete graph and $p = 1$, ℓ_1 collapses to ordinary Euclidean distance. For $p > 1$, paths through dense regions are favored because they can be decomposed into many short edges.

gflow Interpretation

Fermat/PWSPD should be treated as both:

1. a well-studied comparator for density-aware graph geodesics;
2. a special implicit-density case of $d_{G,\phi,\alpha}$, where no explicit density estimate is used and density enters through local sample spacing.

When reporting Fermat results, gflow should state the convention, p , the graph used for computation, any rescaling, and the implied $\alpha = (p-1)/\hat{d}$ if an intrinsic dimension estimate is available.

6. Density-Aware Neighbor Selection

Density-aware distance can enter before edge weighting by changing graph topology. Let $r_k(i)$ be the distance from x_i to its k th nearest neighbor. A kNN-radius density surrogate is

$$\hat{\rho}_i \propto \frac{k}{n r_k(i)^{\hat{d}}},$$

where $\hat{d}$ is an intrinsic dimension estimate.

Possible topology rules include:

1. choose smaller effective k_i in dense regions and larger k_i in sparse regions, subject to a maximum edge length;
2. admit an edge only if it satisfies both a geometric length rule and a density-support rule;
3. connect mutual-kNN components using bridges ranked by geometric length and density support;
4. prune edges whose midpoint or edge-neighborhood density is too low.

These rules do not themselves define a distance. They define G , after which a graph metric is obtained from edge weights.

7. Diffusion Distance

Definition

Given a kernel K_ε , define a density estimate

$$q_i = \sum_j K_\varepsilon(i, j).$$

Diffusion-map alpha normalization uses

$$K_a(i, j) = \frac{K_\varepsilon(i, j)}{q_i^a q_j^a}, \quad P_a(i, j) = \frac{K_a(i, j)}{\sum_k K_a(i, k)}.$$

The diffusion distance at time t is

$$D_t^2(i, j) = \sum_k \frac{(P_a^t(i, k) - P_a^t(j, k))^2}{\pi_k},$$

where π is the stationary or reference measure used by the diffusion construction.

Density Role

The parameter a controls sampling-density normalization:

$a = 0$ keeps strong density drift,

$a = 1$ removes sampling-density drift in the continuum limit.

This a is unrelated to the gflow density exponent α .

Metric Status

Diffusion distance is a random-walk distributional distance. It is a metric on states with distinct transition profiles under the chosen time and weighting; it becomes a pseudometric if two points have identical diffusion profiles or if the embedding truncates discriminating eigenvectors. It is not a shortest-path geodesic.

gflow Interpretation

Diffusion distance should be a comparator or diagnostic family. It answers: “Do two points diffuse to similar neighborhoods?” It does not answer exactly: “What is the cheapest supported path between two points?”

8. PHATE Potential Distance

PHATE constructs a Markov operator P , chooses diffusion time t , forms $P_t = P^t$, transforms rows of P_t into potentials, then compares those potentials.

A simplified definition is

$$U_\gamma(P_t(i, \cdot)) = \begin{cases} \log P_t(i, \cdot), & \gamma = 1, \\ \text{power or square-root-like potential}, & \gamma \approx 0, \end{cases}$$

and

$$D_{\text{PHATE}}(i, j) = \|U_\gamma(P_t(i, \cdot)) - U_\gamma(P_t(j, \cdot))\|_2.$$

PHATE is density-adjacent because high-density paths affect transition probabilities, but the effect is mediated by graph construction, kernel normalization, diffusion time, and the potential transform.

Metric status: PHATE potential distance is an embedding/potential distance, usually a pseudometric if potential vectors coincide or if approximations collapse points. It should not be labeled a geodesic distance.

9. Resistance, Commute, Heat-Kernel, and PageRank Distances

Resistance and Commute

For graph Laplacian L , effective resistance is

$$R_{ij} = (e_i - e_j)^T L^+ (e_i - e_j).$$

Commute time satisfies

$$C_{ij} = \text{vol}(G) R_{ij}.$$

Resistance is a graph metric on connected undirected weighted graphs with positive conductances. It rewards many parallel high-conductance paths. It is not a shortest-path metric.

Heat Kernel

The graph heat kernel is

$$H_t = \exp(-tL).$$

Heat-kernel distances compare rows of H_t . On a smooth manifold, the small-time heat kernel recovers geodesic distance through Varadhan's formula:

$$d_g(x, y)^2 = -\lim_{t \rightarrow 0} 4t \log H_t(x, y).$$

Away from this limit, heat distances are diffusion-scale distances rather than geodesic distances.

Personalized PageRank

Personalized PageRank from source i solves

$$r_i = \beta e_i + (1 - \beta)r_i P.$$

It is generally an asymmetric proximity. Symmetrized transformations may be useful, but they are not automatically metrics.

10. Directed and Gradient-Flow Variants

For trajectory or gradient-flow analysis, `gflow` may eventually need asymmetric path costs. A generic directed edge weight is

$$w_{ij}^{\rightarrow} = \frac{\ell_{ij}}{\phi_{ij}^{\alpha}} + \lambda B_{ij},$$

where B_{ij} penalizes movement against a vector field, potential, time arrow, or biological gradient. The resulting shortest-path distance

$$d^{\rightarrow}(i, j) = \min_{\pi: i \rightarrow j} \sum_{(u,v) \in \pi} w_{uv}^{\rightarrow}$$

is generally not symmetric. It should be called a directed distance or quasi-metric, not a metric.

This is out of scope for the first density-aware implementation but should be kept conceptually separate from symmetric density-aware geodesics.

11. Density-Aware Oracle Truth

For synthetic data with known density ρ_{true} , define oracle truth by

$$d_{\text{oracle}, \alpha}(x, y) = \inf_{\gamma: x \rightarrow y} \int_{\gamma} \rho_{\text{true}}(\gamma(s))^{-\alpha} ds.$$

In a dense oracle graph with oracle nodes z_a , use

$$w_{ab}^{\text{oracle}} = \frac{\|z_a - z_b\|}{\rho_{\text{true}}(m_{ab})^{\alpha}},$$

where m_{ab} is a midpoint or edge-density quadrature point.

Oracle distances are not estimators; they define the benchmark target for a chosen scientific interpretation of support. Reports must state α , density floor, support truncation, grid resolution, and sample-attachment rule.

12. Recommended `gflow` Definition Set

For the next implementation and benchmark phase, use these names:

1. **density-weighted continuous geodesic:** $d_{\rho, \alpha}$, defined by $\int \rho^{-\alpha} ds$.
2. **density-weighted graph geodesic:** $d_{G, \phi, \alpha}$, shortest paths with $w_{ij} = \ell_{ij} / \phi_{ij}^{\alpha}$.

3. **Fermat/PWSPD distance**: shortest paths with edge powers ℓ_{ij}^p , reported with convention and implied $\alpha = (p - 1)/\hat{d}$.
4. **diffusion distance**: row distance between P_a^t transition distributions.
5. **PHATE potential distance**: distance between transformed diffusion-potential rows.
6. **resistance/commute/heat/PageRank distances**: random-walk or operator distances, useful analogies and comparators but not density-weighted geodesics.
7. **density-aware oracle distance**: synthetic truth metric using known density and explicit support rules.

13. Minimal Diagnostics Required

Every report using $d_{G,\phi,\alpha}$, Fermat/PWSPD, or an oracle density-weighted distance should include:

1. density estimate distribution and transform;
2. floor/cap counts and fractions;
3. edge length versus density-weighted edge length;
4. density summary used for each edge;
5. sensitivity over α or p ;
6. sample paths through dense and sparse regions;
7. underestimation and overestimation path attribution;
8. comparison to Euclidean, unweighted graph geodesic, Fermat/PWSPD, diffusion, and PHATE distances when available;
9. tail and outlier influence diagnostics;
10. stability across seeds and sample sizes.

14. Boundary Between Definition and Implementation

The definitions above do not decide:

1. whether density is estimated in raw, PCA, CLR/Aitchison, latent, graph, or oracle space;
2. which graph constructor is best;
3. which edge density summary is best;
4. which default α , floor, or cap should be used;
5. whether density should be exploited or corrected away for a given biology problem.

Those questions belong to the implementation design memo and benchmark reports. This memo only fixes the mathematical language that those reports should use.